

Computer Networks

Micro Project | 2026–27

A. LAN WITH MULTIPLE SWITCHES

Design & Simulation using Cisco Packet Tracer

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Video Link :-

<https://drive.google.com/drive/folders/1b4pealyJ-oxbkN38IyzaCFMJuUrgapTn?usp=sharing>

Department Of Computer Engineering

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1. Aim And Objectives

Aim

To design and simulate a Local Area Network (LAN) using multiple Cisco 2960 switches and PCs in Cisco Packet Tracer and to verify communication between all connected devices.

Objectives

- To understand the basic concept of LAN.
 - To understand the working of a network switch.
 - To connect multiple switches together.
 - To connect multiple PCs using switches.
 - To configure IPv4 addresses on PCs.
 - To understand MAC address learning in switches.
 - To verify IP configuration using the ipconfig command.
 - To verify switch port status.
 - To test communication using the ping command.
 - To understand how multiple switches can extend a LAN.
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2. Introduction

2.1 What is a LAN?

A Local Area Network (LAN) is a computer network that connects computers and other devices in a small geographical area such as a computer laboratory, classroom, office or building.

A LAN allows connected devices to share information and communicate with each other.

2.2 What is a Network Switch?

A network switch is a Layer 2 networking device used to connect multiple devices in a LAN.

A switch receives Ethernet frames and forwards them to the correct port using MAC addresses.

2.3 Why Multiple Switches?

A single switch has a limited number of ports. Multiple switches can be connected together to:

- Connect more computers.
- Increase the number of available ports.
- Extend the physical network.
- Connect different groups of computers.
- Make the network easy to expand.

2.4 MAC Address Learning

Every network device has a MAC address. A switch learns the source MAC address of a received frame and stores it in its MAC address table.

When the switch knows the destination MAC address, it can forward the frame to the correct port.

MAC address learning allows the switches to efficiently forward Ethernet frames to the appropriate destination port.

3. Network Requirement

3.1 Hardware / Packet Tracer Devices

Device	Quantity	Purpose
Cisco 2960-24TT Switch	3	LAN switching
PC-PT	6	End devices
Copper Straight-Through Cable	6	PC-to-switch connections
Copper Cross-Over Cable	2	Switch-to-switch connections
Cisco Packet Tracer	1	Network simulation

3.2 Software

Cisco Packet Tracer

4. Network Topology

The proposed LAN consists of three Cisco 2960-24TT switches and six PCs. Switch0 is used as the central switch, while Switch1 and Switch2 are used as access switches. Switch1 and Switch2 are connected to Switch0. Two PCs are connected to each switch. All PCs belong to the same IPv4 network, 192.168.10.0/24.

4.1 Network Structure

Switch	Role	Connected PCs
Switch0	Central Switch	PC4, PC5
Switch1	Access Switch	PC0, PC1
Switch2	Access Switch	PC2, PC3

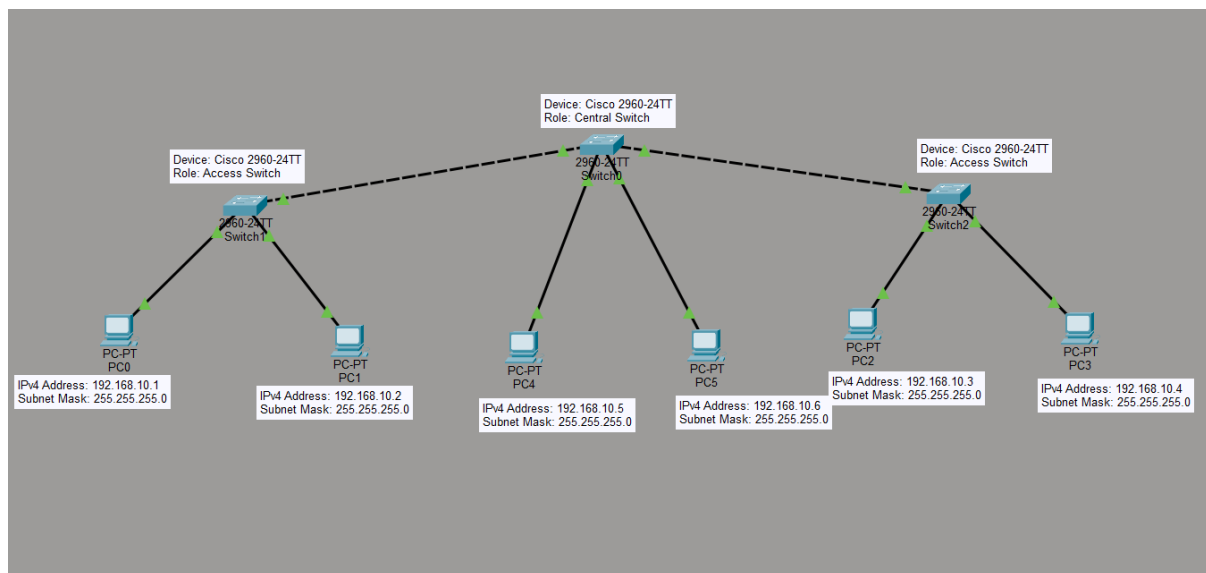


Figure 1: Cisco Packet Tracer topology of LAN with multiple switches

5. IP Addressing Scheme

All PCs are configured in the same network:

Network: 192.168.10.0/24

Subnet Mask: 255.255.255.0

IP Address Table

Device	IP Address	Subnet Mask	Network
PC0	192.168.10.1	255.255.255.0	192.168.10.0/24
PC1	192.168.10.2	255.255.255.0	192.168.10.0/24

PC2	192.168.10.3	255.255.255.0	192.168.10.0/24
PC3	192.168.10.4	255.255.255.0	192.168.10.0/24
PC4	192.168.10.5	255.255.255.0	192.168.10.0/24
PC5	192.168.10.6	255.255.255.0	192.168.10.0/24

Default Gateway

A default gateway is not required because all PCs are in the same IP network.

5.1 IP Address Configuration

The IP address is configured manually on each PC.

Steps

1. Open a PC.
2. Click **Desktop**.
3. Click **IP Configuration**.
4. Select **Static**.
5. Enter the IP address.
6. Enter subnet mask 255.255.255.0.
7. Do not enter a default gateway.

Example — PC0

IP Address: 192.168.10.1
Subnet Mask: 255.255.255.0
Default Gateway: blank

Repeat the same process for PC1 to PC5.

The screenshot shows a Windows Command Prompt window on PC0. The window title is 'PC0'. The tabs at the top are 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The Command Prompt shows the output of the 'ipconfig' command. It displays the configuration for 'FastEthernet0 Connection: (default port)' and 'Bluetooth Connection'. The FastEthernet0 configuration shows a Link-local IPv6 Address of FE80::230:F2FF:FED9:65E3, an IPv6 Address of ::, an IPv4 Address of 192.168.10.1, a Subnet Mask of 255.255.255.0, and a Default Gateway of 0.0.0.0. The Bluetooth connection shows a Link-local IPv6 Address of ::, an IPv6 Address of ::, an IPv4 Address of 0.0.0.0, a Subnet Mask of 0.0.0.0, and a Default Gateway of 0.0.0.0.

```

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FED9:65E3
    IPv6 Address . . . . . : ::
    IPv4 Address . . . . . : 192.168.10.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . . : ::
    IPv4 Address . . . . . : 0.0.0.0
    Subnet Mask . . . . . : 0.0.0.0
    Default Gateway . . . . . : ::
                                0.0.0.0

```

Figure 2: PC0 IPv4 configuration verified using ipconfig

6. Devices And Port Configuration

6.1 Device Configuration

Device	Model	Role	Connected PCs
Switch0	Cisco 2960-24TT	Central Switch	PC4, PC5
Switch1	Cisco 2960-24TT	Access Switch	PC0, PC1
Switch2	Cisco 2960-24TT	Access Switch	PC2, PC3

6.2 Switch Port Allocation

Switch	Port	Connected Device	Function
Switch0	Fa0/1	PC4	End Device
Switch0	Fa0/2	PC5	End Device

Switch0	Fa0/23	Switch1 Fa0/23	Inter-Switch Link
Switch0	Fa0/24	Switch2 Fa0/23	Inter-Switch Link
Switch1	Fa0/1	PC0	End Device
Switch1	Fa0/2	PC1	End Device
Switch1	Fa0/23	Switch0 Fa0/23	Uplink
Switch2	Fa0/1	PC2	End Device
Switch2	Fa0/2	PC3	End Device
Switch2	Fa0/23	Switch0 Fa0/24	Uplink

6.3 Connecting The Devices

PC-to-Switch Connections

Connect:

PC0 → Switch1 Fa0/1

PC1 → Switch1 Fa0/2

PC2 → Switch2 Fa0/1

PC3 → Switch2 Fa0/2

PC4 → Switch0 Fa0/1

PC5 → Switch0 Fa0/2

Use:

Copper Straight-Through Cable

Switch-to-Switch Connections

Connect:

Switch1 Fa0/23 → Switch0 Fa0/23

Switch2 Fa0/23 → Switch0 Fa0/24

Use:

Copper Cross-Over Cable

7. Switch Verification

The switches are used as Layer 2 devices in this LAN. No IP address configuration is required on the switch for basic PC-to-PC communication. The operation of the switches is verified using Cisco IOS commands.

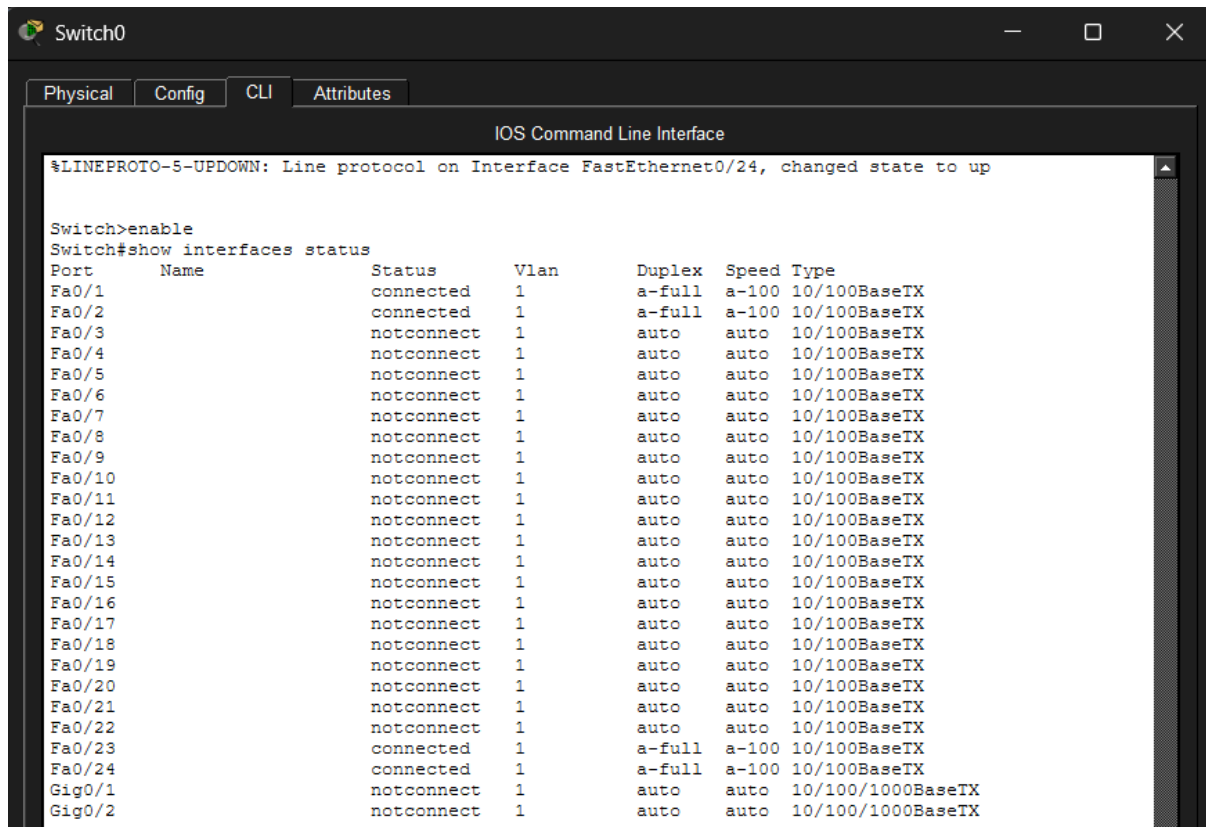
7.1 Verify Port Status

Open **Switch0** → **CLI**.

Enter:

```
enable
show interfaces status
```

This command displays the status of switch ports.



The screenshot shows the CLI of a switch named Switch0. The command 'show interfaces status' has been executed, displaying a table of port statuses. The table has columns for Port, Name, Status, Vlan, Duplex, Speed, and Type. Ports Fa0/1 and Fa0/23 are connected, while all other ports are in a 'notconnect' state. The output also includes a message about a line protocol change on Fa0/24.

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch>enable
Switch#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1         a-full  a-100  10/100BaseTX
Fa0/2     Fa0/2     connected   1         a-full  a-100  10/100BaseTX
Fa0/3     Fa0/3     notconnect  1         auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1         auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1         auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  1         auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  1         auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1         auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1         auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1         auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1         auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1         auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto   10/100BaseTX
Fa0/22    Fa0/22    notconnect  1         auto    auto   10/100BaseTX
Fa0/23    Fa0/23    connected   1         a-full  a-100  10/100BaseTX
Fa0/24    Fa0/24    connected   1         a-full  a-100  10/100BaseTX
Gig0/1    Gig0/1    notconnect  1         auto    auto   10/100/1000BaseTX
Gig0/2    Gig0/2    notconnect  1         auto    auto   10/100/1000BaseTX
```

Figure 3: Switch0 Port Status Verification

7.2 Mac Address Table Verification

A switch automatically learns the MAC addresses of connected devices.

To check the MAC address table:

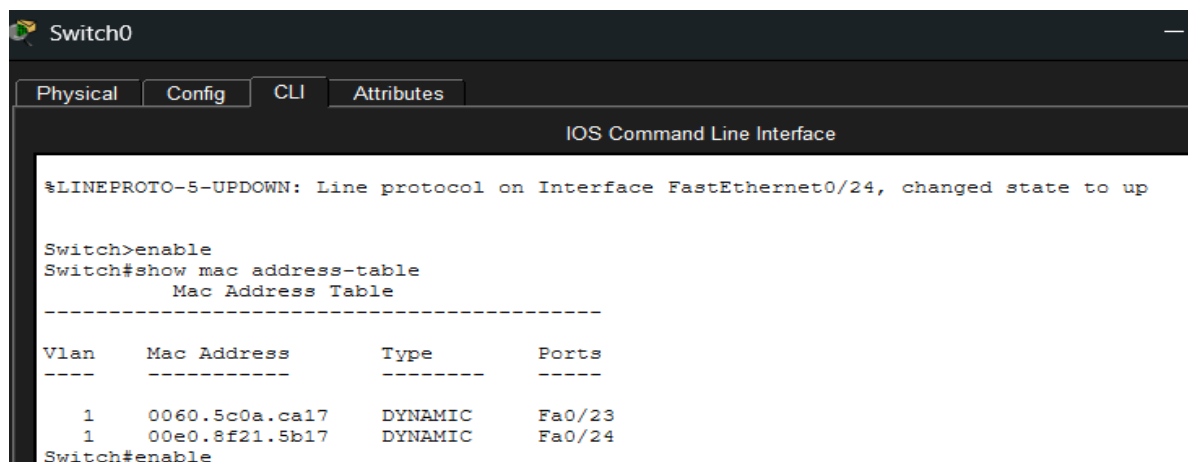
enable

show mac address-table

The output displays:

- VLAN
- MAC Address
- Type
- Port

The MAC address table confirms that the switch has learned devices connected to its ports.

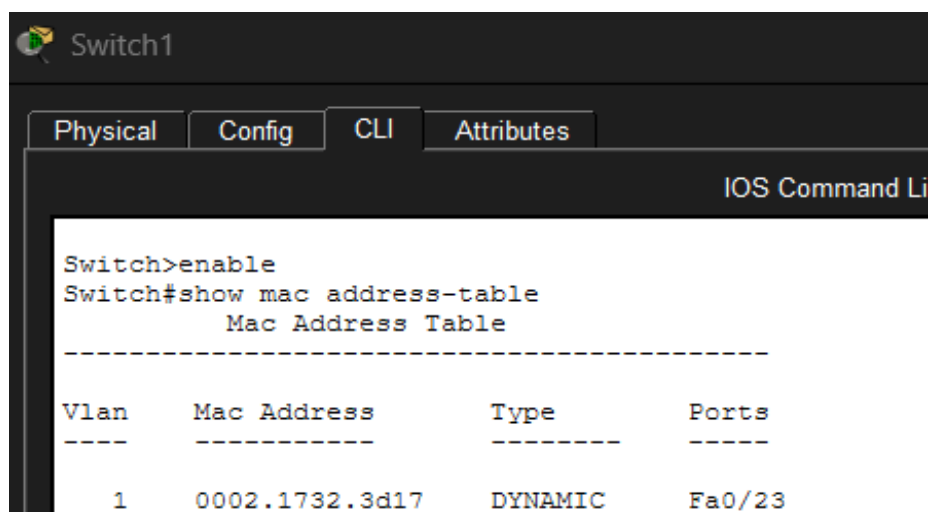


```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0060.5c0a.ca17   DYNAMIC     Fa0/23
1       00e0.8f21.5b17   DYNAMIC     Fa0/24
Switch#enable
```

Figure 4: Switch0 MAC address table



```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface

Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0002.1732.3d17   DYNAMIC     Fa0/23
```

Figure 5: Switch1 MAC address table

8. Network Connectivity Testing

After configuring the IP addresses and connecting all devices, network connectivity was tested using the ping command.

The ping command is used to check whether one device can communicate with another device through the network.

Steps

1. Open **PC0**.
2. Click **Desktop**.
3. Open **Command Prompt**.
4. Enter the ping command.
5. Test communication with different PCs.

Commands Used

```
ping 192.168.10.2  
ping 192.168.10.3  
ping 192.168.10.5  
ping 192.168.10.6
```

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
```

Figure 6: Successful ping tests from PC0

8.1 Ping Test Results

The following ping tests were performed from PC0:

Source PC	Destination PC	Result	Packet Loss
PC0	PC1 (192.168.10.2)	Successful	0%
PC0	PC2 (192.168.10.3)	Successful	0%
PC0	PC4 (192.168.10.5)	Successful	0%
PC0	PC5 (192.168.10.6)	Successful	0%

Result

All tested PCs successfully communicated with PC0. Each ping test received 4 replies and showed 0% packet loss. This confirms that the devices connected through the multiple switches can communicate successfully.

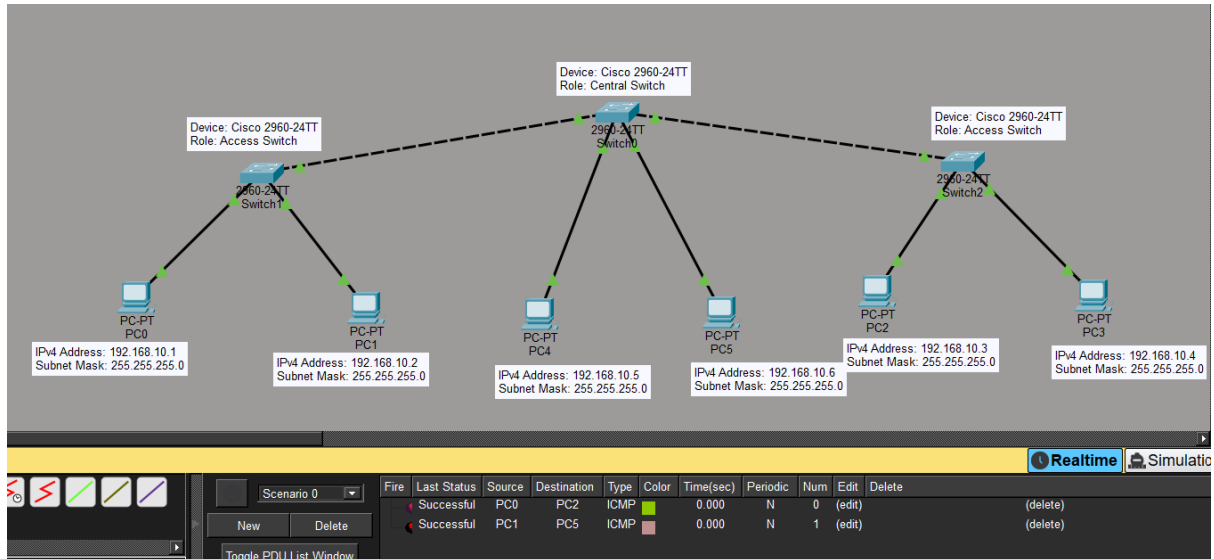


Figure 7: Successful ICMP communication in Simulation Mode

9. Result Analysis

The network was tested after completing the configuration.

The following results were observed:

- All three switches were connected successfully.
- All six PCs were connected to the switches.
- All PCs were configured in the same network.
- Switch ports showed a connected status.
- MAC addresses were learned by the switches.
- Ping communication between different PCs was successful.
- The ping test showed 0% packet loss.

Result Table

Test	Expected Result	Actual Result
PC IP Configuration	Correct IP address	Successful
Switch Port Status	Connected	Successful
MAC Address Learning	MAC addresses learned	Successful
PC-to-PC Ping	Reply received	Successful
Packet Loss	0%	0%

Final Result

The LAN with multiple switches was successfully designed and simulated in Cisco Packet Tracer.

10. Working Of The Network

The working of the network is as follows:

1. PC0 and PC1 are connected to Switch1.
2. PC2 and PC3 are connected to Switch2.
3. PC4 and PC5 are connected to Switch0.
4. Switch1 and Switch2 are connected to the central Switch0.
5. All PCs use the same network 192.168.10.0/24.
6. When a PC sends data, the switch receives the Ethernet frame.
7. The switch learns the source MAC address.
8. The switch uses its MAC address table to forward frames.
9. The frame travels through the required switch-to-switch link when the destination PC is connected to another switch.
10. The destination PC receives the data.
11. Successful ping replies confirm that the network is working.

Simple Network Flow

```
PC0
↓
Switch1
↓
Switch0
↓
Switch2
↓
PC2
```

For example, when PC0 communicates with PC2, the data travels through the switches and reaches PC2.

11. Advantages

- Easy to connect more computers.
- Provides more network ports.
- Easy to expand the LAN.
- Provides reliable communication.
- Easy to manage.
- Suitable for small and medium networks.
- Multiple switches can cover a larger area.

12. Applications

A LAN using multiple switches can be used in:

- Computer laboratories.
 - College networks.
 - School networks.
 - Offices.
 - Small businesses.
 - Department networks.
 - Small organizations.
-

13. Limitations

- This project uses a single IP network.
 - No router is used.
 - Communication outside the LAN is not demonstrated.
 - No VLAN configuration is used.
 - IP addresses are configured manually.
 - The project is designed for a basic LAN demonstration.
-

14. Future Scope

The network can be improved in the future by:

- Adding more switches.
 - Adding more PCs.
 - Creating different VLANs.
 - Adding a router.
 - Using DHCP for automatic IP configuration.
 - Connecting the LAN to other networks.
 - Adding network security features.
-

15. Conclusion

This project successfully demonstrates a LAN using three Cisco 2960 switches and six PCs in Cisco Packet Tracer. All PCs were configured with IPv4 addresses in the 192.168.10.0/24 network. Switch connectivity, port status, and MAC address learning were successfully verified. Connectivity was tested using ping with 0% packet loss between the PCs. The project provided practical understanding of LAN, switches, IP addressing, MAC addresses, and network communication.

B. Implementation of Web Server and DNS Server Network

using Cisco Packet Tracer

1. Aim

To design and implement a small client-server network using **Cisco Packet Tracer**, where three PCs access a website hosted on a Web Server using a domain name resolved by a DNS Server.

2. Objectives

1. To create a LAN using a Cisco switch.
 2. To connect three client PCs to the network.
 3. To configure a Web Server.
 4. To configure a DNS Server.
 5. To assign IP addresses to all devices.
 6. To map a domain name with the Web Server IP using DNS.
 7. To access the website from multiple PCs using the domain name.
 8. To verify successful communication between clients, DNS Server and Web Server.
-

3. Software / Tools Required

- Cisco Packet Tracer
 - 1 × Cisco 2960 Switch
 - 3 × PCs
 - 1 × Web Server
 - 1 × DNS Server
 - Copper Straight-Through cables
-

4. IP Addressing Table

Device	IP Address	Subnet Mask
PC0	192.168.10.1	255.255.255.0
PC1	192.168.10.2	255.255.255.0
PC2	192.168.10.3	255.255.255.0
Web Server	192.168.10.4	255.255.255.0
DNS Server	192.168.10.5	255.255.255.0

All devices are connected to the same LAN.

5. Network Topology

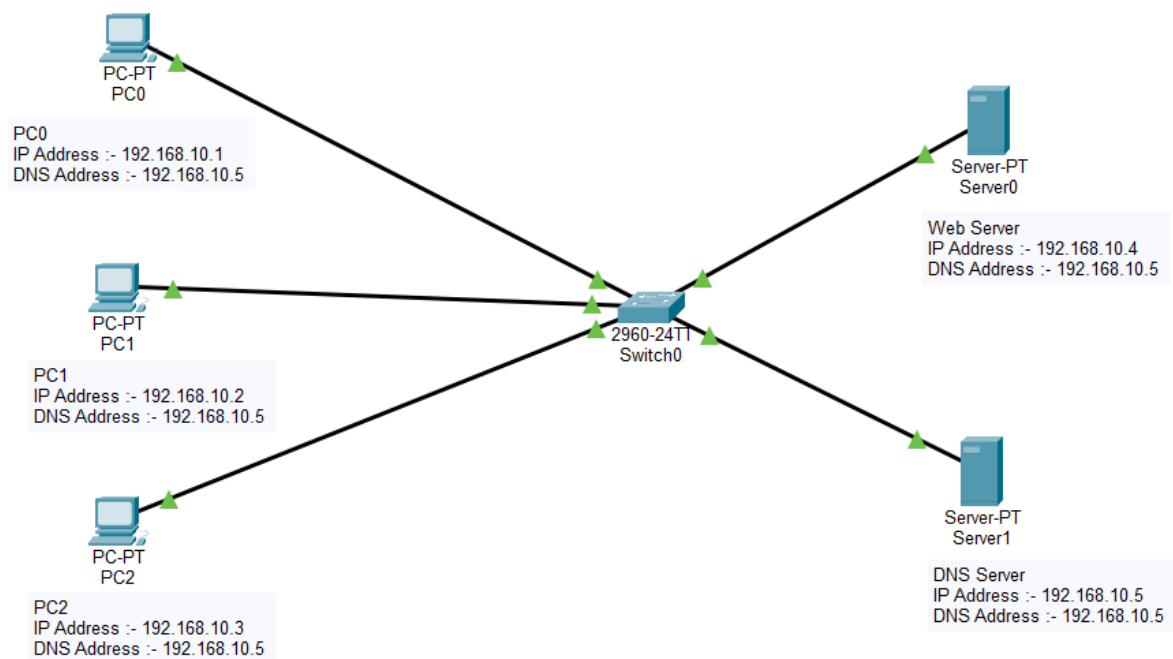


Figure 1: Network topology of Web Server and DNS Server using Cisco Packet Tracer

6. Procedure

Step 1: Create the Network

Open Cisco Packet Tracer.

Place the following devices:

- 3 PCs
- 1 Cisco 2960 switch
- 1 Web Server
- 1 DNS Server

Connect all devices to the switch using **Copper Straight-Through** cables.

Switch connections:

Device	Switch Port
PC0	Fa0/1
PC1	Fa0/2
PC2	Fa0/3
Web Server	Fa0/4
DNS Server	Fa0/5

7. Configure IP Address of PC0/PC1/PC2

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.10.1

Subnet Mask: 255.255.255.0

DNS Server: 192.168.10.5

Do that for PC1 And PC2.

IP Addressing Table :

Device	IP Address	Subnet Mask	DNS Server
PC0	192.168.10.1	255.255.255.0	192.168.10.5
PC1	192.168.10.2	255.255.255.0	192.168.10.5
PC2	192.168.10.3	255.255.255.0	192.168.10.5

8. Configure Web Server

Open:

Web Server → Desktop → IP Configuration

Enter:

IP Address: 192.168.10.4

Subnet Mask: 255.255.255.0

Then go to:

Services → HTTP

Make sure HTTP is ON.

And then, edit index.html that default webpage, for example:

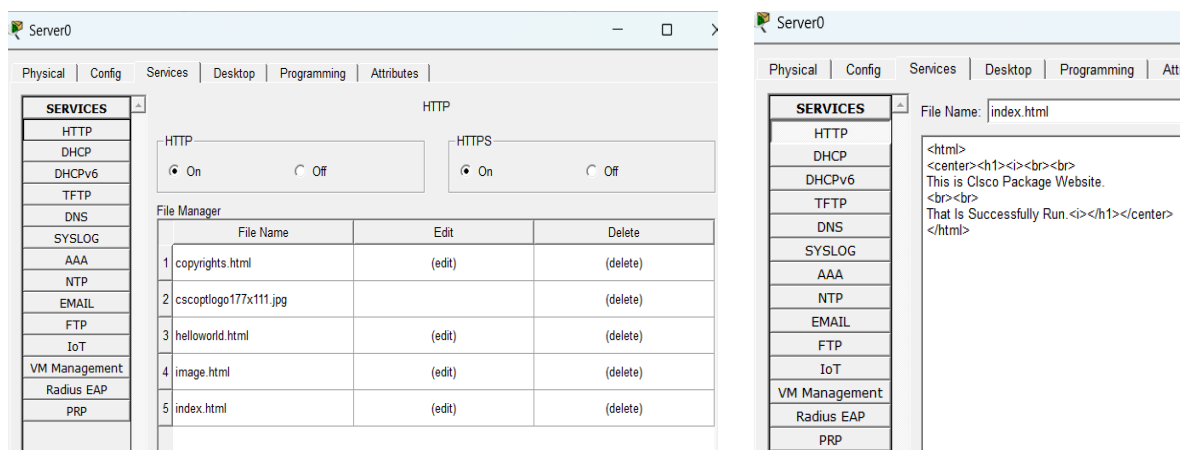


Figure 2: Web Server configuration / Service http

This website is hosted on a Cisco Packet Tracer Web Server.

9. Configure DNS Server

Open:

DNS Server → Desktop → IP Configuration

Enter:

IP Address: 192.168.10.5

Subnet Mask: 255.255.255.0

Then go to:

Services → DNS

Turn **DNS Service ON**.

Add the following DNS record for example:

Name: www.hrweb.com

Address: 192.168.10.4

Click **Add**.

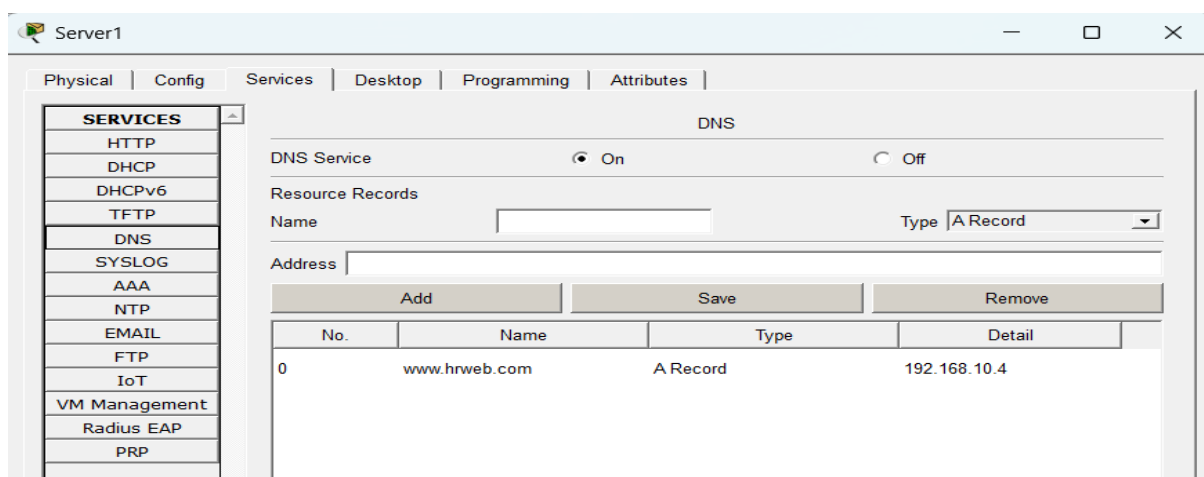


Figure 3: DNS server configuration

10. Test DNS Resolution

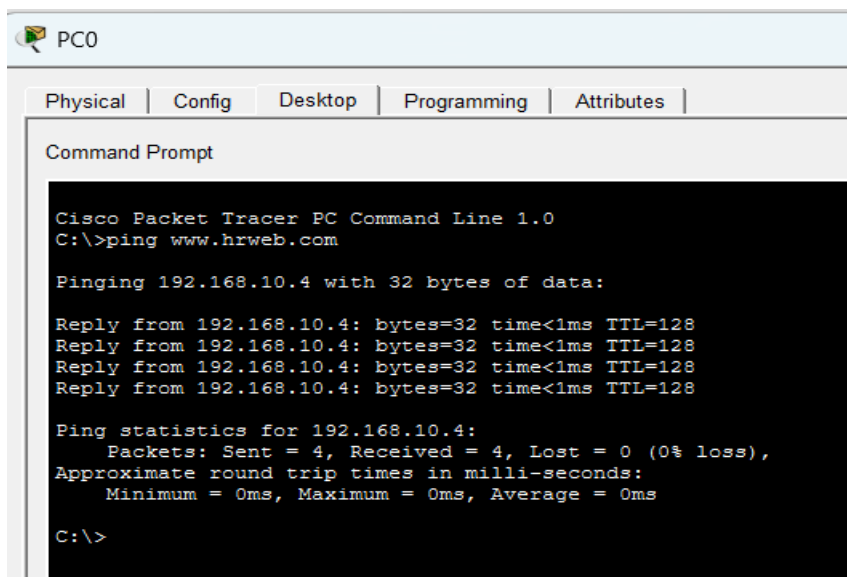
From PC0:

Desktop → Command Prompt

Enter:

ping www.hrweb.com

The DNS Server should convert www.hrweb.com to 192.168.10.4

A screenshot of the PC0 Command Prompt window in Cisco Packet Tracer. The window title is 'PC0' and it has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The command prompt text is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.hrweb.com

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 4: DNS name resolution test

11. Access Website from PC0

Open:

PC0 → Desktop → Web Browser

Enter:

http://www.hrweb.com

The website hosted on the Web Server should open.

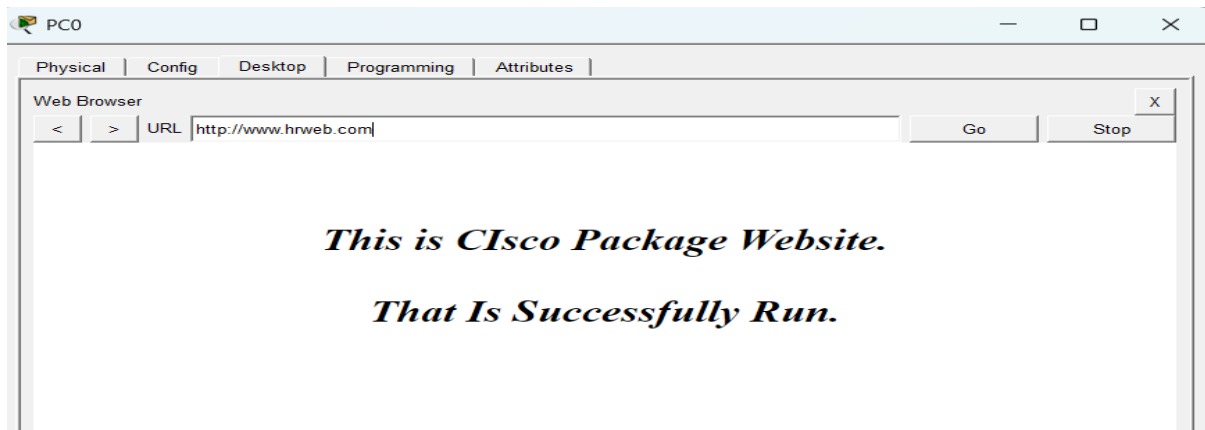


Figure 5: Website accessed from PC0 using domain name

12. Access Website from PC1

Open the Web Browser on PC1 and enter:

`http://www.hrweb.com`

The same website should open.

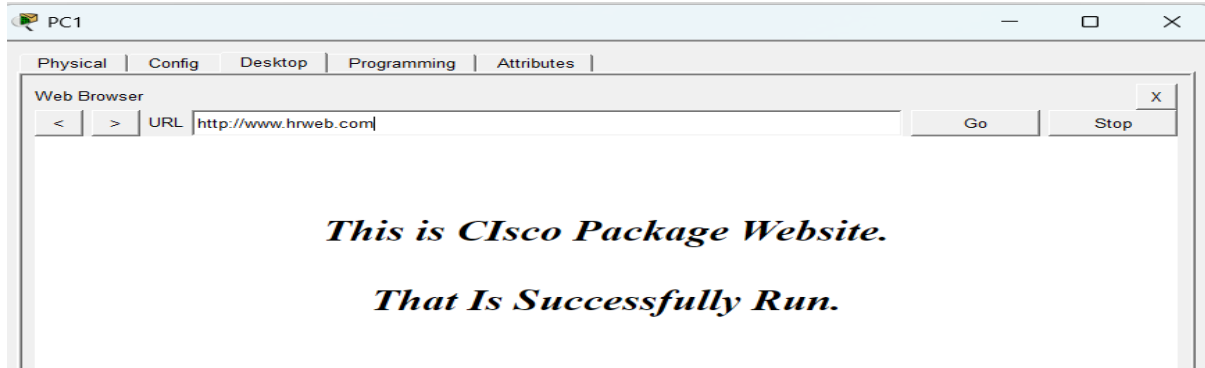


Figure 6: Website accessed from PC1

13. Access Website from PC2

Open the Web Browser on PC2 and enter:

<http://www.hrweb.com>

The same website should open.

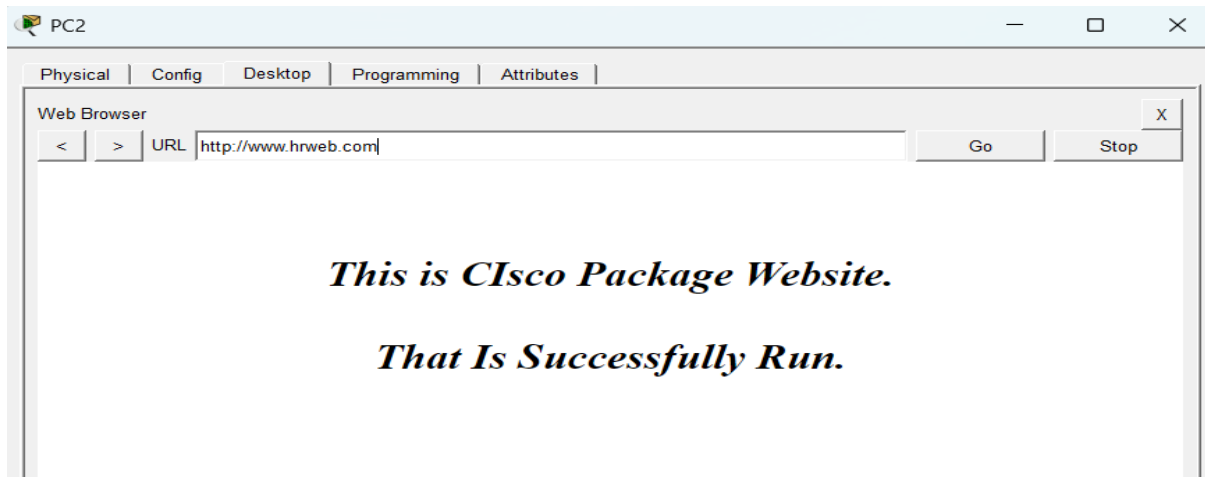


Figure 7: Website accessed from PC2

14. Working of the Project

The user enters the domain name www.hrweb.com in the web browser of PC0, PC1, or PC2. The request is sent through the switch to the DNS Server. The DNS Server checks the configured DNS record and converts the domain name www.hrweb.com into the IP address 192.168.10.4, which is the IP address of the Web Server. After receiving the Web Server IP address, the client PC sends a request to the Web Server through the switch. The Web Server processes the request and sends the requested webpage back to the client PC. In this way, PC0, PC1, and PC2 can access the same website using the domain name instead of directly entering the Web Server IP address.

Explanation

1. The client PC requests the IP address of www.hrweb.com.
2. The request reaches the DNS Server through the switch.
3. The DNS Server checks its DNS record.
4. DNS returns 192.168.10.4.
5. The client contacts the Web Server at 192.168.10.4.
6. The Web Server sends the webpage to the client.
7. The same process can be performed by PC0, PC1 and PC2.

15. Result

The Web Server and DNS Server were successfully configured in Cisco Packet Tracer.

Three PCs were connected to the same LAN and were able to access the website using the domain name:

www.hrweb.com

Therefore, the implementation of a **Web Server and DNS Server Network with multiple clients** was successfully completed.

16. Advantages

- Multiple PCs can access the same Web Server.
- DNS makes website access easier using a domain name.
- The network is simple and easy to manage.
- It demonstrates client-server communication.
- It can be expanded by adding more clients and servers.

17. Conclusion

The project demonstrates how a small LAN can provide web services to multiple client computers. The DNS Server converts the domain name into the Web Server IP address, while the Web Server provides the webpage to the clients.

The project successfully demonstrates **IP addressing, switching, DNS resolution, Web Server configuration and client-server communication** using Cisco Packet Tracer.